

Research Discussion Assignment 1

Course: DATA 612 - Recommender Systems

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Part 1: YouTube as a Commercial Recommender System

For this discussion, I selected YouTube as a commercial recommender system. YouTube is one of the most widely used recommendation platforms in the world and is responsible for suggesting videos based on a user's interests and viewing history. I use YouTube regularly for learning technology topics, watching tutorials, and entertainment content.

I believe YouTube uses a combination of content-based filtering and collaborative filtering. Content-based filtering recommends videos that are similar to videos I have watched before. For example, if I watch Azure, Python, or cybersecurity tutorials, YouTube will recommend more videos on those topics. Collaborative filtering looks at users with similar interests and recommends videos that those users enjoyed. By combining both approaches, YouTube is able to provide personalized recommendations that are usually relevant to each user.

Does YouTube Deliver a Good Experience?

In my experience, YouTube provides a very good recommendation experience. Most of the videos recommended on my homepage are related to topics that interest me, such as technology, cloud computing, cybersecurity, and programming. However, the system is not perfect. Sometimes YouTube recommends too many videos on the same topic, which can limit content discovery. Overall, I think YouTube's recommendations are accurate and useful most of the time.

Part 2: Attacks on Recommender Systems

Summary of the Washington Post Article

The Washington Post article describes how the movie *The Promise* was targeted by a coordinated online campaign before its release. Thousands of users gave the movie very low ratings on IMDb without actually watching it because the film addressed the Armenian Genocide, a topic that remains politically controversial. This type of attack, known as review bombing, artificially lowered the movie's rating and demonstrated how recommender and rating systems can be manipulated by organized groups. The article highlights the

challenges of maintaining fair and trustworthy recommendation systems when users intentionally submit misleading reviews.

Other Examples of Recommender System Attacks

There are many other examples of attacks on recommender systems. One common example is fake product reviews on Amazon, where sellers create fake accounts or pay people to leave positive reviews to boost product rankings. Another example is review bombing of movies, TV shows, or video games when groups of users intentionally leave extremely low ratings due to political or social disagreements rather than the actual quality of the content. These actions can mislead users and reduce the reliability of recommendation systems.

Preventing Attacks on Recommender Systems

Recommender systems can be protected by implementing measures that reduce fraudulent activity. For example, platforms can require verified purchases before allowing product reviews, limit ratings from newly created accounts, and use machine learning techniques to detect suspicious review patterns. In the case of movies and TV shows, platforms could prevent users from rating content until after it has been released. These measures can help improve the accuracy, fairness, and trustworthiness of recommendation systems.